

The analysis of TTC delay data was conducted to identify relationships between delays and factors such as weather, time, and location. The findings suggest that delays are primarily concentrated along two major subway lines and are not significantly influenced by external factors such as weather.

Where do delays happen?

Based on both the number and total duration of delays, twelve stations stand out as the most affected. As shown in the map below, these stations are located along two main corridors, one vertical and one horizontal, corresponding to the routes of Line 1 (Yonge–University) and Line 2 (Bloor–Danforth).

The size of each point represents the number of delays, while the color intensity reflects the total delay duration.

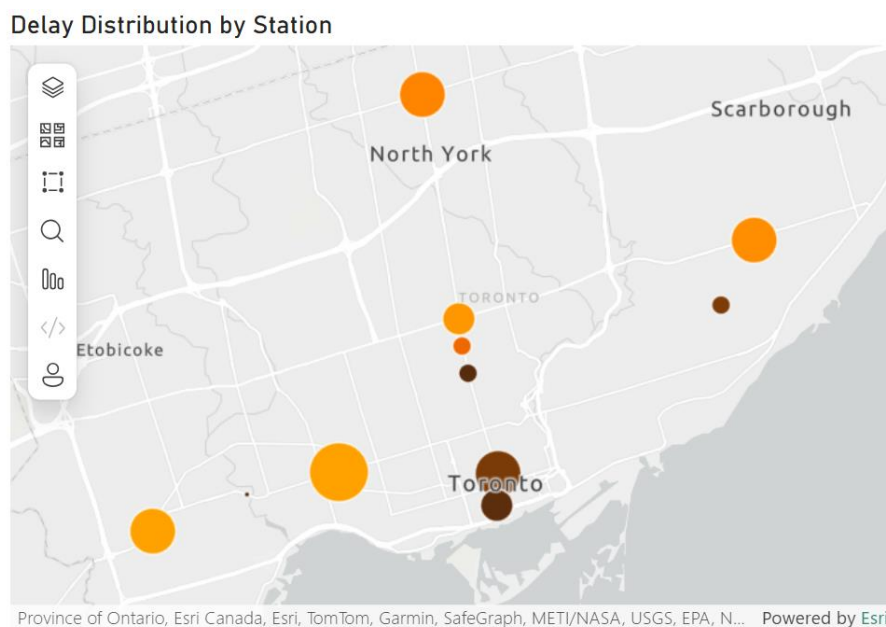


Figure 1 Distribution of delays across TTC stations, highlighting concentration along Line 1 and Line 2.

Comparing delay counts across lines confirms this observation quantitatively, with the majority of delays occurring on Line 1 and Line 2.

Line	Sum of Min_Delay	Count of Delay
Line 1	42375	14525
Line 2	29163	10984
Other	2728	897
Grand Total	74266	26406

When do delays happen?

There are only minor differences in the monthly and weekly distribution of delays. However, more meaningful insights are observed when analyzing delays by hour of the day.

Figure 2 shows the number of delays and total delay duration by hour. While delays are relatively consistent throughout the day, slight increases are observed during peak commuting hours.

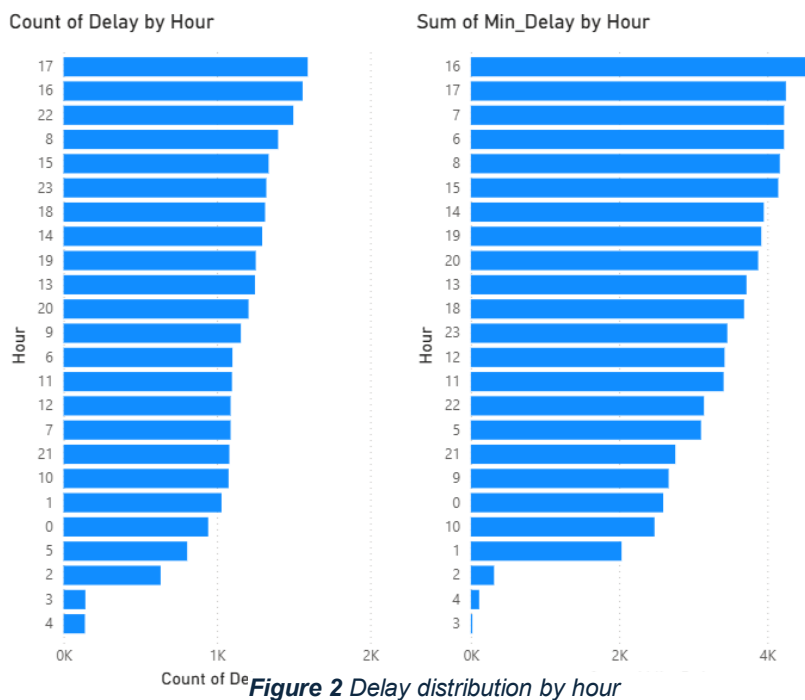


Figure 2 Delay distribution by hour

Additionally, a significant increase in delays occurs during late evening hours, particularly around 10 PM and 11 PM, suggesting that the issue extends beyond traditional peak periods (Figure 3).

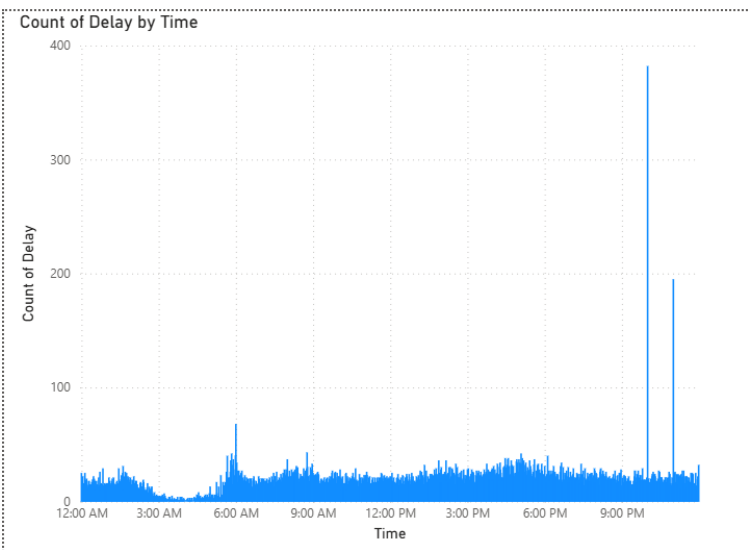


Figure 3 Number of Delays by Time

Why do delays happen?

Weather analysis shows little to no correlation between delay volume and total precipitation, indicating that weather is not a primary driver of delays.

Instead, the analysis suggests that delays are more closely related to internal factors, such as high passenger demand and congestion along key transit lines. The increase in delays during peak hours, as well as the unexpected late-evening peaks, supports this conclusion.

The TTC subway network structure below helps to better understand the observed delay patterns.

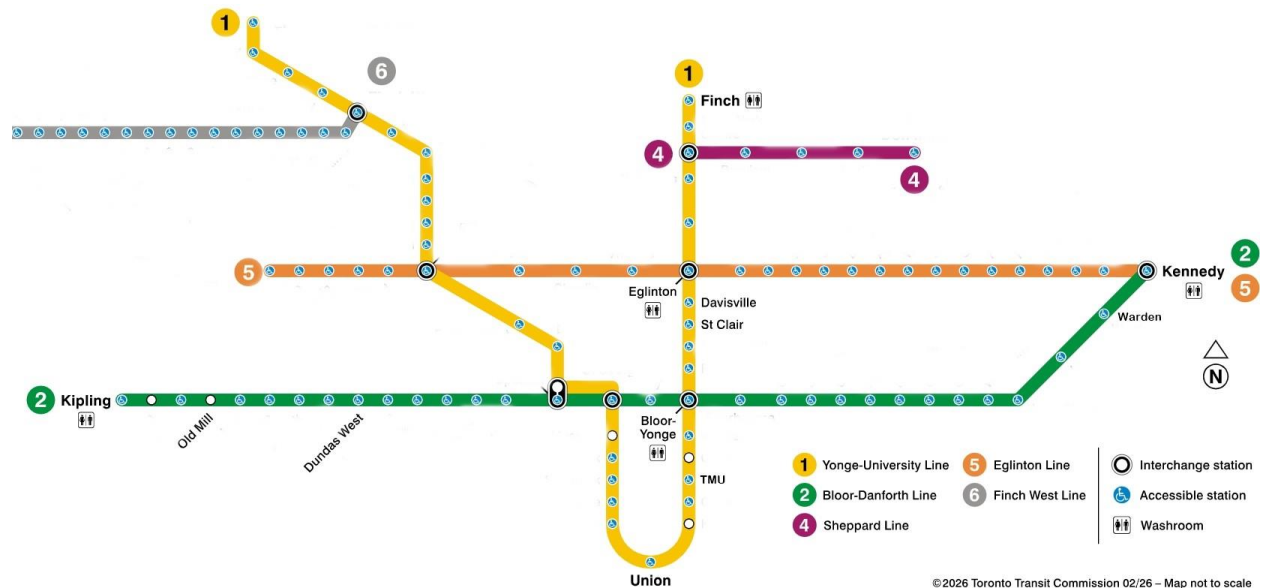


Figure 4 TTC subway system map showing stations with the largest volume of delays

Solutions

While TTC schedules already provide frequent service during peak hours, the observed delay patterns suggest that additional adjustments may be beneficial.

It is recommended to further increase service frequency during peak hours (4 PM–6 PM) for lines 1 and 2, and to evaluate the need for more frequent service during late evening hours (10 PM–11 PM). This could help reduce congestion.

Additionally, it is recommended to allocate more resources to high-delay stations, including increased staff presence, in order to reduce and prevent delays from spreading across the network.

Can this model be replicated?

This model can be applied to other transit systems or industries where operations depend on location and time-based scheduling. Such areas include the airline industry, logistic / delivery services, etc.